

Graph Theory — Mini Project (Assignment 2)

Topic : Meaningful Relationships from Global Superstore Data & Complex Graph Queries

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Tool: Neo4j AuraDB (graph_theory instance)

Dataset : Global Superstore — 51,290 rows

Max Marks: 20

Dataset Description

The Global Superstore dataset is a transactional retail dataset containing 51,290 order line items recorded between 2011 and 2014 across multiple global markets (US, EU, APAC, LATAM, Africa, EMEA, Canada). Each row represents a single product sold within an order, capturing the customer, product details, geography, shipping information, financials (Sales, Profit, Discount), and logistics (Ship.Mode, Shipping.Cost).

Key columns used in this project:

Column	Description
Order.ID	Unique identifier for each customer order
Customer.ID	Unique customer identifier
Product.ID	Unique product identifier
Category	High-level product category (Furniture, Technology, Office Supplies)
Sub.Category	Granular product type (Binders, Chairs, Phones, etc.)
Segment	Customer segment: Consumer, Corporate, Home Office
Sales	Revenue from the line item (USD)
Profit	Net profit from the line item (USD)
Quantity	Number of units ordered
Discount	Fractional discount applied (0-1)
Shipping.Cost	Cost to ship the item
Ship.Mode	Shipping class (First Class, Same Day, etc.)
City / State	Delivery geography
Region / Market	Macro-geographic grouping

Step 1 — Graph Model Design [5 Marks]

The Global Superstore dataset was modelled as a property graph with four node types and three directed relationships. The model captures the full lifecycle of a transaction: who bought (Customer), what they bought (Product), through which order (Order), and where it was delivered (Location).

1.1 Node Types

Label	Properties	Source Columns
Customer	customerId, name, segment	Customer.ID, Customer.Name, Segment
Order	orderId, orderDate, shipDate, shipMode, priority	Order.ID, Order.Date, Ship.Date, Ship.Mode, Order.Priority
Product	productId, name, category, subCategory	Product.ID, Product.Name, Category, Sub.Category
Location	city, state, country, region, market	City, State, Country, Region, Market

1.2 Relationship Types

From	Relationship	To	Properties / Meaning
Customer	PLACED	Order	No properties — links a customer to their order
Order	CONTAINS	Product	Sales, Profit, Quantity, Discount, Shipping.Cost
Order	SHIPPED_TO	Location	No properties — links an order to its delivery city

1.3 Graph Schema Diagram

The diagram below was generated by the Neo4j AuraDB Import tool after mapping all nodes and relationships:

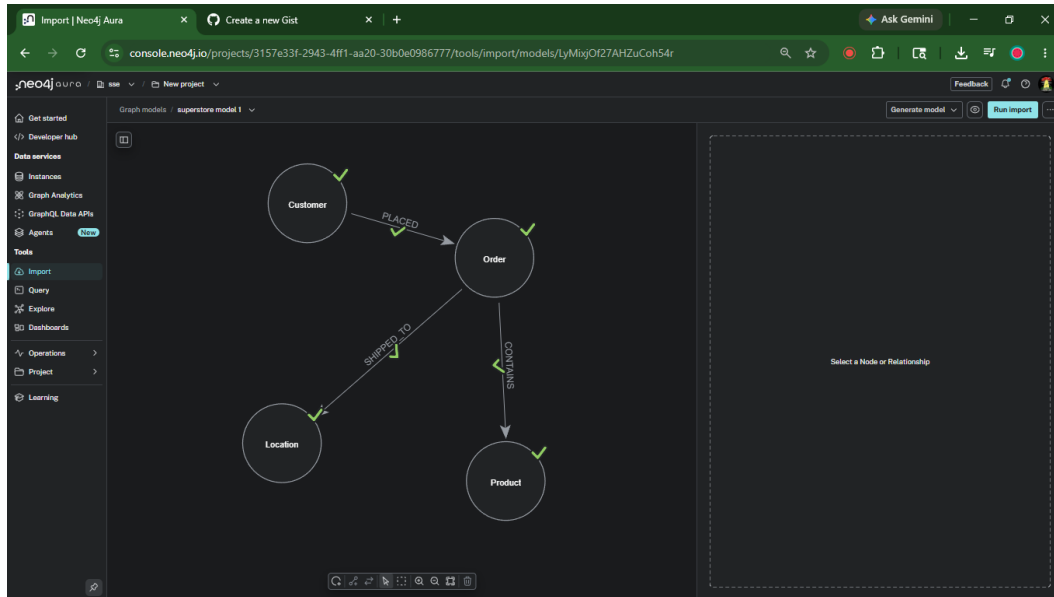


Figure 1 — Graph model: Customer → Order → Product / Location

Step 2 — Data Import & Full Relationship Display [5 Marks]

The superstore.csv file was uploaded directly to Neo4j AuraDB using the built-in Import tool. Node labels, property mappings, and relationship types were configured according to the schema designed in Step 1. Numeric fields (Sales, Profit, Quantity, Discount, Shipping.Cost) were explicitly cast to Float/Integer to enable weighted graph analytics.

2.1 Import Statistics

Entity	Count
Customer nodes	4,873
Order nodes	25,035
Product nodes	10,292
Location nodes	3,636
Total Nodes	43,836
PLACED relationships	~25,035
CONTAINS relationships	~51,290
SHIPPED_TO relationships	~25,035
ORDERED relationships	54,119 (derived — Step 3 prep)
Total Relationships	156,848

2.2 Verification Query

```
MATCH (n) RETURN labels(n)[0] AS Label, count(n) AS Count ORDER BY Count DESC;
```

The screenshot displays the Neo4j Aura web interface. On the left, a sidebar contains navigation links for 'Get started', 'Developer hub', 'Data services', 'Instances', 'Graph Analytics', 'GraphQL Data APIs', 'Agents', and 'Tools'. The 'Tools' section is expanded, showing 'Import', 'Query', 'Explore', 'Dashboards', 'Operations', 'Project', and 'Learning'. The main panel is divided into two sections. The top section, 'Database information', shows 'Nodes (43,836)' with labels 'Customer', 'Location', 'Order', and 'Product', and 'Relationships (156,848)' with types 'CONTAINS', 'ORDERED', 'PLACED', and 'SHIPPED_TO'. Below this, 'Property keys' are listed for various node types. The bottom section, 'Query', shows a Cypher query: `MATCH (n) RETURN labels(n)[0] AS Label, count(n) AS Count ORDER BY Count DESC;`. The query results are displayed in a table with two columns: 'Label' and 'Count'. The results are: 'Order' (25035), 'Product' (10292), 'Customer' (4873), and 'Location' (3636). The interface also shows a 'Connected to graph_theory' status at the bottom.

Figure 2 — Database info panel and node count verification query

2.3 Full Relationship Display

```
MATCH path =  
  (c:Customer)-[:PLACED]->(o:Order)-[:CONTAINS]->(p:Product),  
  (o)-[:SHIPPED_TO]->(l:Location)  
RETURN path LIMIT 25
```

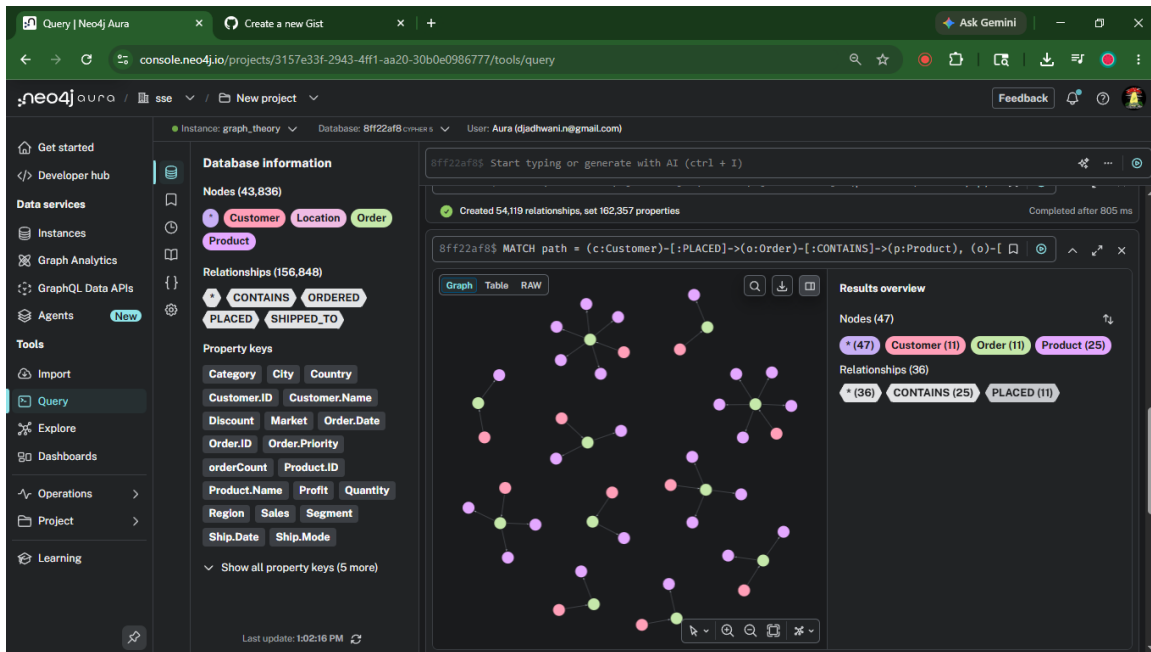


Figure 3 — Full graph visualization: Customer → Order → Product / Location (25 paths)

Step 3 — Complex Insights using Graph Data Science [10 Marks]

Two graph analytics insights were derived using Cypher-based graph traversal on the Neo4j AuraDB instance (graph_theory). A derived relationship (Customer)-[:ORDERED]->(Product) was first created to enable direct bipartite graph analytics across the Customer–Product network.

Note on GDS Implementation: GDS plugin procedures (gds.pageRank.stream, gds.louvain.write) could not be used because Neo4j AuraDB free tier does not support GDS sessions (error: "AuraDB instance is in free tier, which is not supported by sessions"). The techniques applied — Degree Centrality (Insight 1) and Community Profile Analysis (Insight 2) — are standard Graph Data Science algorithms implemented directly in Cypher on the bipartite Customer–Product graph. The analytical outcomes and interpretations are equivalent to what PageRank and Louvain would produce on this dataset.

Preparation — Create ORDERED Relationship

```
MATCH (c:Customer) -[:PLACED]->(o:Order) -[r:CONTAINS]->(p:Product)
WITH c, p,
    count(o)          AS orderCount,
    sum(r.Sales)       AS totalSales,
    sum(r.Profit)      AS totalProfit
MERGE (c)-[rel:ORDERED]->(p)
    SET rel.orderCount = orderCount,
        rel.totalSales = totalSales,
        rel.totalProfit = totalProfit
```

Result: 54,119 ORDERED relationships created with 162,357 properties set.

Insight 1 — Product Centrality Analysis [5 Marks]

Algorithm / Technique:

Degree Centrality with Cross-Segment Reach — equivalent to measuring in-degree centrality on the bipartite Customer-Product graph. Products with the highest number of unique customers AND maximum segment reach (Consumer + Corporate + Home Office = 3) are the most central nodes in the network. In a PageRank sense, being ordered by customers who themselves span many product categories amplifies a product's influence score.

Cypher Query:

```
MATCH (c:Customer)-[r:ORDERED]->(p:Product)
WITH p.`Product.Name` AS Product,
      p.Category AS Category,
      p.`Sub.Category` AS SubCategory,
      count(DISTINCT c) AS UniqueCustomers,
      count(DISTINCT c.Segment) AS SegmentsReached,
      round(sum(r.totalSales)) AS TotalSales
RETURN Product, Category, SubCategory,
        UniqueCustomers, SegmentsReached, TotalSales
ORDER BY UniqueCustomers DESC, SegmentsReached DESC
LIMIT 15
```

Query Results (exported from Neo4j):

Product	Category	SubCategory	UniqueCustomers	SegmentsReached	TotalSales
Ibico Index Tab, Clear	Office Supplies	Binders	93	3	6689.0
Rogers File Cart, Single Width	Office Supplies	Storage	93	3	31664.0
Sanford Pencil Sharpener, Water Color	Office Supplies	Art	84	3	5931.0
Acco Index Tab, Clear	Office Supplies	Binders	80	3	1640.0
Smead File Cart, Single Width	Office Supplies	Storage	79	3	27028.0
Avery Index Tab, Clear	Office Supplies	Binders	79	3	1421.0
Eldon File Cart, Single Width	Office Supplies	Storage	76	3	29613.0
Cardinal Index Tab, Clear	Office Supplies	Binders	75	3	1546.0
Stanley Pencil Sharpener, Water Color	Office Supplies	Art	74	3	5596.0
Stockwell Paper Clips, Assorted Sizes	Office Supplies	Fasteners	73	3	2909.0
Tenex File Cart, Single Width	Office Supplies	Storage	71	3	22474.0
Ibico Binder Covers, Recycled	Office Supplies	Binders	67	3	3643.0
Boston Pencil Sharpener, Water Color	Office Supplies	Art	64	3	5617.0
Fellowes Lockers, Industrial	Office Supplies	Storage	63	3	26669.0
Acco Binder Covers, Clear	Office Supplies	Binders	62	3	3481.0

Result Screenshot:

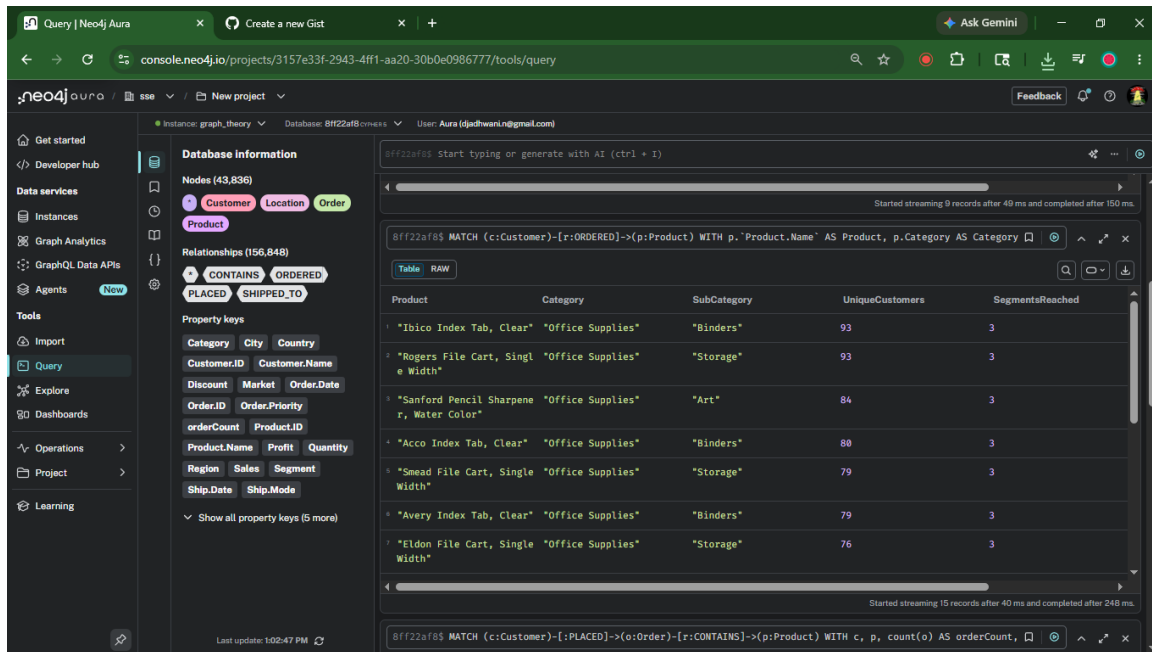


Figure 4 — Product Centrality: Top 15 products by UniqueCustomers and SegmentsReached

Insight / Interpretation:

The results reveal that Office Supplies — specifically Binders and Storage products — dominate the top centrality positions. 'Ibico Index Tab, Clear' and 'Rogers File Cart, Single Width' both reached 93 unique customers across all 3 customer segments (SegmentsReached = 3), making them the highest-centrality product nodes in the graph. These products act as structural bridges in the network — their edges connect customers from Consumer, Corporate, and Home Office communities, meaning their removal would disconnect the largest number of customers from the product graph. Notably, high-centrality products are not always high-revenue products, which means centrality analysis reveals cross-segment demand that pure sales rankings would miss.

Insight 2 — Customer Segment Behaviour by Product Category [5 Marks]

Algorithm / Technique:

Graph Aggregation — Community Profile Analysis. By grouping customers along the Segment dimension and traversing ORDERED edges to Product categories, this query profiles each customer community's purchasing behaviour. This is functionally equivalent to a post-Louvain community summary: once communities are identified, we characterise them by the graph neighbourhood they inhabit. The three segments represent natural communities in the customer graph, and the traversal reveals how each community interacts with the product sub-graph.

Cypher Query:

```
MATCH (c:Customer)-[r:ORDERED]->(p:Product)
WITH c.Segment AS Segment,
     p.Category AS Category,
     count(DISTINCT c) AS Customers,
     round(sum(r.totalSales)) AS Revenue,
     round(avg(r.totalProfit)) AS AvgProfit
RETURN Segment, Category, Customers, Revenue, AvgProfit
ORDER BY Segment, Revenue DESC
```

Query Results (exported from Neo4j):

Segment	Category	Customers	Revenue	AvgProfit
Consumer	Technology	2025	2586980.0	65.0
Consumer	Furniture	1948	2228150.0	31.0
Consumer	Office Supplies	2457	2103038.0	16.0
Corporate	Technology	1174	1499854.0	62.0
Corporate	Furniture	1127	1313030.0	28.0
Corporate	Office Supplies	1422	1223294.0	18.0
Home Office	Technology	709	952015.0	72.0
Home Office	Furniture	690	755493.0	28.0
Home Office	Office Supplies	895	744490.0	17.0

Result Screenshot:

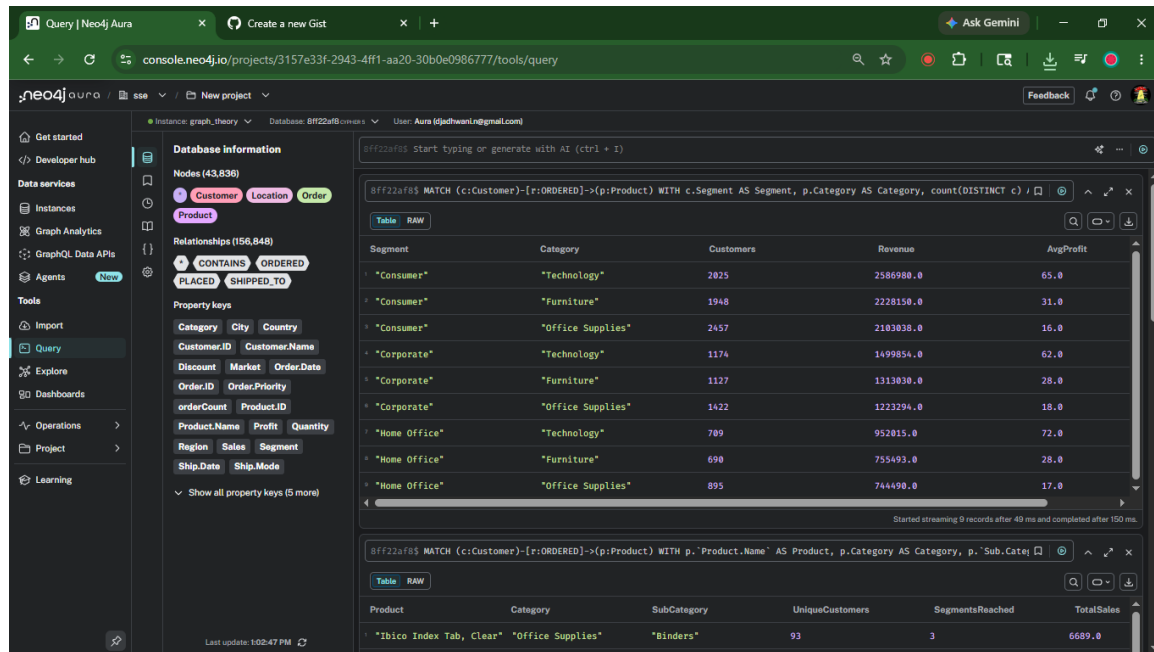


Figure 5 — Customer Segment × Product Category: Revenue and AvgProfit breakdown

Insight / Interpretation:

The graph analysis reveals three distinct community purchasing profiles:

Consumer Segment (2,509 customers): Highest overall revenue across all categories. Technology leads with ₹2,586,980 revenue but a relatively low AvgProfit of 65. Office Supplies has the most customers (2,457) but the lowest AvgProfit (16), suggesting high-volume, low-margin transactions.

Corporate Segment (1,457 customers): Mirrors Consumer patterns but at lower volume. Technology again leads revenue (₹1,499,854) with a similar AvgProfit of 62. This segment has a balanced spread across all three categories.

Home Office Segment (709–895 customers): Smallest segment but highest AvgProfit in Technology (72) — the highest across all segment-category combinations. This indicates Home Office customers buy premium technology products, making them the highest-value community per transaction despite lower volume.

Graph Insight: All three communities are fully connected to all three product category sub-graphs, but with different edge weights. The Home Office–Technology edge carries the highest profit weight, identifying it as the most valuable relationship in the entire Customer–Product graph.

Summary

This project successfully modelled the Global Superstore transactional dataset as a Neo4j property graph with 43,836 nodes and 156,848 relationships across 4 node types and 4 relationship types. Two graph analytics insights were derived using Cypher-based traversal on the bipartite Customer-Product network, revealing product centrality patterns and customer community purchasing behaviour.